

Variational Graph Attention Networks With Self-Supervised Learning for Multivariate Time Series Anomaly Detection

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Abstract—Cyber-physical systems (CPSs) are multidimensional and complex systems that integrate computing, networking, and physical environment, including various sensors that can produce vast multivariate time series data. Recently, scholars have proposed many unsupervised anomaly detection methods to monitor and alarm potential faults of CPSs. However, current methods still have some limitations. First of all, it is challenging to capture spatial-temporal relationships explicitly. In the second place, the anomaly detection model based on the variational autoencoder (VAE) may not use the latent variable and often falls into a kind of collapse, which is encouraged by the Kullback–Leibler (KL) divergence term. In the end, more information about the anomalies is needed. This article proposes a novel anomaly detection method called variational graph attention networks with self-supervised learning (VGATSL). We use stacked spatial-temporal graph attention (STGAT) networks to capture the temporal and feature correlations. Next, a VAE is constructed based on bidirectional gated recurrent units (BiGRUs), and the von Mises-Fisher (vMF) distribution is used as its latent space. Indeed, this distribution places latent representation on the surface of a unit hypersphere to allow for uninformed priors, thus preventing the model collapse. Additionally, self-supervised learning is introduced to simulate abnormal behaviors and to help the model form more explicit normality boundaries. Extensive experiments on three public real-world datasets demonstrate that VGATSL outperforms the baseline methods. We also illustrate that VGATSL has superior interpretability and flexibility.

Index Terms—Latent variable model, multivariate time series, self-supervised learning, spatial-temporal graph attention (STGAT) network, unsupervised anomaly detection.

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I. INTRODUCTION

DUE to the development of informatization, many sensors are deployed in complex cyber-physical systems (CPSs), such as autonomous vehicles and water treatment plants, to produce vast multivariate time series data with different characteristics, attributes, and scales [1]. CPSs are large-scale and geographically dispersed. There may be implicit dependency relationships between sensors. The fault of a particular sensor can spread to other related devices, causing cascading failures of components or the entire system. Usually, detecting abnormal time points from multivariate time series data helps monitor and discover potential risks, threats, and faults of CPSs, avoiding business interruption and even casualties. However, in the real world, anomalies are rare, and annotating data are expensive [2]. Based on this, unsupervised anomaly detection has received widespread attention.

Unsupervised anomaly detection aims to train a robust model with generalized patterns using unlabeled historical normal data. Once this model cannot fit the current observation, it will alarm [3]. Recently, many methods have been developed [4], [5], [6]. Specifically, prediction-based methods focus on predicting multiple continuous values to discover contextual anomalies, such as long- and short-term memory (LSTM) [7] and temporal convolutional networks (TCNs) [8]. Reconstruction-based methods attempt to explore anomalies in the overall distribution of sub-sequences, including generative adversarial networks (GANs) [9] and autoencoders (AEs) [10]. Mainly, AEs use reconstruction error as the threshold. Some promising models have also been proposed, such as variational AE (VAE) [11] and convolutional AE (CAE) [12]. Additionally, transfer learning-based anomaly detection methods have also attracted plenty of scholars to research, such as DAAN [13], AICDA [14], and FS-ADAPT [15]. Nevertheless, these methods cannot capture the coupling relationship between sensors, making it difficult for them to process non-Euclidean data.

Fig. 1 shows an inspired example from the secure Water Treatment System (SWaT) dataset [16]. This example provides four sensor signals. Overall, sensor signals are time-dependent. The value of the next timestamp is closely related to the trend of changes in the historical series. It can be observed that the signal exhibits significant periodicity. Meanwhile,

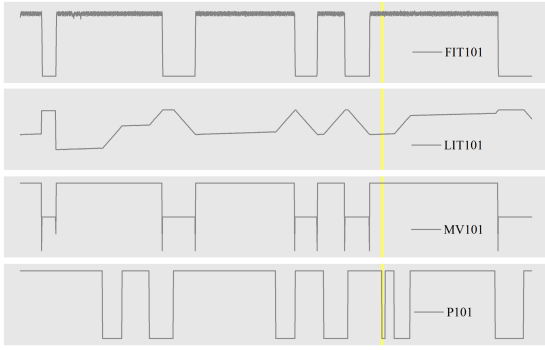


Fig. 1. Illustrative example (from secure water treatment system dataset). The anomaly event is marked as yellow. Four sensors (FIT101, LIT101, MV101, and P101).

Fig. 1 shows an anomaly event (yellow span). This anomaly occurred on sensor LIT101 (set the water level above H) [16]. During this period, the LIT101 value slowly increased, while FIT101 and MV101 remained uninfluenced, and P101 dropped to a low-level value. The above situation fully demonstrates the coupling relationship between LIT101 and P101, and they will interact. In addition, FIT101 and MV101 are not useless, as they can also create an offsetting effect that changes the dependency relationship with other sensors. Therefore, it is challenging to identify anomaly events based on purely temporal patterns. Learning the spatial correlation between multivariate time series is crucial.

Graph neural networks (GNNs) [17] have successfully processed non-Euclidean data. The reason is that GNNs use a message-passing mechanism to extract contextual representations of nodes, which helps to learn spatial features of multivariate time series. For example, graph convolutional network (GCN) [18] has the characteristic of permutation invariance and can use fast localized convolutional filters to process non-Euclidean data. Graph deviation network (GDN) [19] uses attention mechanisms to extract spatial information. Other methods, such as MTAD-GAT [6] and STGAT-MAD [20], can jointly capture spatial-temporal correlations. However, due to limitations of fitting, these methods cannot model the stochastic interrelationships. Recent work has also attempted to combine GNNs with probabilistic methods, considering the randomness of multivariate time series, which helps to reduce intrinsic information loss, such as the graph relational learning networks (GReLeN) [5] and variational graph convolutional recurrent networks (VGCRN) [21]. However, these methods cannot adaptively learn graph structures and often lose discriminative dependencies in the time dimension. Researchers have introduced self-supervised learning into variational GNNs, such as self-supervised variational graph AEs (SS-VGAE) [22], and self-supervised multivariate KPIs anomaly detection (MAD-STA) [23], to improve feature separability or dynamic graph structures. Unfortunately, they cannot access unknown abnormal knowledge and are difficult to optimize. In summary, the research mentioned above has made substantial progress but still faces some critical challenges.

- 1) It is difficult to capture spatial-temporal relationships explicitly. Detection performance will decrease if the implicit information is lost.

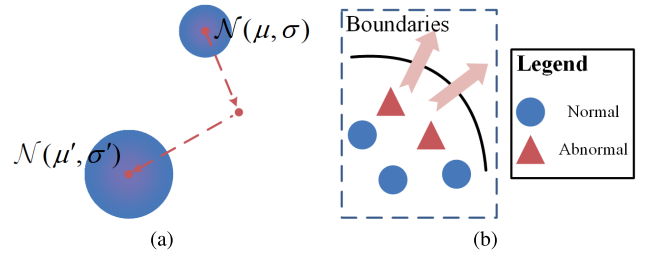


Fig. 2. Visualization of related challenges. (a) If the latent space is a Gaussian distribution, the KL term will force the instance to move toward the prior (centered at the origin). $N(\cdot)$ represents the Gaussian distribution. (b) Lack of abnormal data leads to the model learning biased decision boundaries.

- 2) The VAE-based model is prone to collapse, which will result in a local optimum. Probabilistic methods often assume that the prior follows a Gaussian distribution. Moreover, the Kullback–Leibler (KL) divergence term in the loss function will encourage the posterior toward the prior, which makes model optimization difficult, as indicated in Fig. 2(a).
- 3) The lack of information about the anomalies. For unsupervised anomaly detection, the model's training process relies on unlabeled historical normal data, which makes it unable to access abnormal knowledge and then form inaccurate normality boundaries, as indicated in Fig. 2(b).

To address the abovementioned challenges, we propose a novel variational graph attention network with self-supervised learning (VGATSL) for unsupervised anomaly detection. First, VGATSL adopts a graph feature extractor to capture the temporal and feature representations used for reconstruction. This feature extractor consists of stacked spatial-temporal graph attention (STGAT) networks. In the second place, VGATSL uses the von Mises-Fisher (vMF) distribution as a substitute. vMF distribution places latent representation on the surface of a unit hypersphere. So, KL divergence can be fixed as a constant, structurally preventing collapse and making the model more robust. Last, VGATSL introduces self-supervised learning to simulate abnormal behaviors, which attempts to perturb the normal data to synthesize compelling anomalies for the model.

To summarize, the main contributions of this article are as follows.

- 1) We propose a novel unsupervised anomaly detection method, which utilizes the graph feature extractor to learn spatial-temporal information and builds a VAE based on bidirectional gated recurrent units (BiGRUs) to achieve the reconstruction task.
- 2) We utilize the vMF distribution instead of the Gaussian distribution as latent space, which places the latent representation on the surface of a unit hypersphere to allow for uninformed priors and thus avoid KL collapse.
- 3) We introduce self-supervised learning to address the lack of information about anomalies. Specifically, this module can generate different but realistic abnormal instances to help the model obtain rich original information and learn more explicit normality boundaries.

- 4) Extensive experiments have been carried out on three public real-world datasets to prove the effectiveness and progressiveness of VGATSL.

This article is organized as follows. Section II introduces the related work. Section III presents the preliminaries. Section IV describes VGATSL in detail. Section V reports the experimental results and discussions. Section VI presents the limitations and future work. Section VII draws the conclusions.

II. RELATED WORK

We review the vital anomaly detection research for multivariate time series. Meanwhile, we discuss transfer learning-based anomaly detection to help readers better understand current research hotspots. Considering that our method relies on spherical latent space representation and self-supervised learning, we summarize relevant work on these topics.

A. Multivariate Time Series Anomaly Detection

Traditional methods are classified into statistical learning (e.g., ARIMA [24]), clustering-based (e.g., OCSVM [25]), density-based (e.g., LOF [26]), and isolation-based methods (e.g., iForest [27]). Thanks to its powerful and robust feature extraction capabilities, more and more unsupervised detection tasks are adopting the deep learning paradigm.

Prediction-based methods attempt to observe the current data point and alert for anomalies if its predicted value deviates from the actual value. Recurrent neural networks (RNNs) and convolutional neural networks (CNNs) are the core architectures of prediction-based models. Hundman et al. [28], proposed LSTM-NDT, which utilizes LSTM to achieve accurate predictions and uses a nonparametric thresholding approach to evaluate residuals. This method has been validated in detecting spacecraft anomalies. Kravchik and Shabtai [29] proposed an anomaly detection scheme based on the statistical deviation. Its core architecture is a different variant of CNNs. Lai et al. [30] proposed the LSTNet, which uses CNNs or RNNs to extract time series' local dependencies and trend information.

Reconstruction-based methods focus on compressing data into implicit representations and using the reconstruction error as the detection threshold. Currently, AE is the most popular method [10]. For example, Su et al. [1] developed a robust anomaly detection scheme, which uses the stochastic variables connection technique to learn the normal patterns and then uses reconstruction probabilities to locate anomalies. Additionally, GANs have also been introduced [9]. MAD-GAN [3] sets detection thresholds based on the discrimination and reconstruction errors. Chen et al. [31] proposed DAEMON, which uses adversarial strategies to regularize implicit variables and reconstructed data for robust learning of normal patterns. It is worth mentioning that other methods, such as the deep autoencoding gaussian mixture model (DAGMM) [32], denoising AE (DAE) [33], and Deep convolutional autoencoding memory networks (CAE-M) [34], have also achieved good results.

However, the above methods cannot capture the complex topological relationships between sensors, making it challenging to process non-Euclidean data with potential correlations.

Hence, this article attempts to use graph learning to address this problem.

B. Transfer Learning-Based Anomaly Detection

Transfer learning is dedicated to transferring knowledge from one field to another related field. It is widely applied in anomaly detection tasks like machine monitoring, predictive maintenance, and energy management. Without loss of generality, Transfer learning-based anomaly detection methods mainly include instance transfer, parameter transfer, mapping transfer, and domain-adversarial transfer [35], [36].

Intuitively, instance transfer assigns different weights to source domain data. For example, Vincent et al. [37] selected labeled instances with similar localized data distributions in the source and target domains and then transferred them to the target anomaly detection task. Parameter transfer starts with pretrained model parameters and fine-tunes on the target task. For example, Wen and Keyes [38] proposed a CNN-based transfer learning framework for anomaly detection. This framework pretrains the model on synthetic normal data and then fine-tunes its weights on unknown univariate or multivariate abnormal data.

The basic idea of mapping transfer is to learn the feature representations of the target domain and minimize the distribution differences between the source and the target domains. Typical measuring methods include Wasserstein distance [39], maximum mean discrepancy (MMD) [40], maximum mean square discrepancy [41], and variance discrepancy representation [42]. Qian et al. [42] proposed a vibration characteristic-guided distribution alignment metric, which improves MMD. It can express variance information and enhance discrepancy representation ability. Cao et al. [43] utilized generalized normality learning to minimize the distribution gap between in-distributed and out-of-distribution normal samples. Wang et al. [44] proposed an anomaly-aware contrastive alignment model to align contrastive representations and labeled normal representations. Qian et al. [14] proposed an adaptive intermediate class-wise distribution alignment model, which aligns the global and class-wise distributions of different domains and adaptively achieves domain confusion.

Inspired by GANs [9], domain-adversarial transfer aims to extract generalized features through adversarial training. For example, Mao et al. [13] proposed a novel deep one-class Transfer learning method based on domain adversarial training, which adaptively transfers one-class detection rules to improve end-to-end anomaly detection performance. Li et al. [15] proposed a Few-shot anomaly detection framework, which uses a dueling triplet network to address the constraints of unsupervised target information and learns invariant features by adversarial training.

In summary, transfer learning-based anomaly detection methods have attracted plenty of scholars to research. We review these well-known works to help readers better understand current research hotspots.

C. Graph Learning in Multivariate Time Series

GNN has succeeded in modeling multivariate time series [45], [46], [47], [48]. For example, GDN [19] adopts

a flexible framework to construct graphs and uses attention mechanisms to encode information. MTAD-GAT [6] treats all timestamps and variables as different complete graphs and jointly optimizes them. STGAT-MAD [20] uses multiscale STGAT networks to capture spatial-temporal correlations. These methods cannot model the stochastic interrelationships during reconstruction, considering the limited fitting. Researchers attempted to combine probabilistic methods with GNNs. Chen et al. [21], proposed VGCRN, which models the channel dependence and randomness using the graph convolutional recurrent network (GCRN). GReLeN [5] combined GNNs and stochastic relation learning mechanisms to capture sensor interdependence and established a composite metric. These methods have achieved impressive results, but they cannot learn the evolutionary patterns of graph structures well and may lose discriminative dependencies in time dimension. Based on this, researchers introduced self-supervised learning into variational GNNs. For example, Zhang et al. [22] proposed a SS-VGAE method. It searches for the optimal center of the latent space, enhancing the separability of mean features. Chen et al. [23] proposed a self-supervised multivariate anomaly detection method that combines adaptive graph structure learning with improved graph attention networks (GAT). Hojjati et al. [49] used self-supervised learning to map the adjacent partitions into similar representations, which helps the model learn discriminative features. However, these anomaly detection methods based on SS-VGAEs still cannot access abnormal knowledge and are prone to forming inaccurate normality boundaries. In this article, we use self-supervised learning to simulate abnormal behaviors, which helps to synthesize compelling anomalies for the model. This is significantly different from the previous research. Additionally, existing variational graph AEs typically assume that the prior follows a Gaussian distribution, which makes model optimization difficult and causes mode collapse. The proposed method uses the vMF distribution as latent space, which places the latent representation on the surface of a unit hypersphere to allow for uninformed priors and thus avoid KL collapse. It is worth mentioning that recent research has also introduced theories such as discrete wavelet transform and adversarial learning into GNNs [50], [51], [52], [53].

D. Self-Supervised Learning

Researchers are committed to developing self-supervised multivariate time series methods suitable for downstream tasks [54], [55], [56], [57], [58], [59], [60]. For example, Eldele et al. [58] transformed the original time series data into multiple views and used temporal and contextual contrastive learning to capture robust representations. Yue et al. [59] proposed TS2Vec, which adopts a hierarchical mechanism to perform contrastive learning and utilizes an aggregation operation to capture the sub-sequence representation. Nevertheless, these pretext tasks still cannot provide information related to anomalies, making it difficult to simulate real-world abnormal behaviors. Xu et al. [60], used a series of perturbation operations to generate abnormal instances. These instances help the model to obtain rich abnormal information. Indeed, this

method can address the lack of knowledge about anomalies and promote the model to learn more explicit normality boundaries.

E. Spherical Latent Space

Probabilistic generative models (PGMs) have achieved outstanding results in image classification, text translation, sequence prediction, and other fields [61], [62], [63]. PGMs often assume that the prior of the latent space follows a Gaussian distribution. This may lead to collapse. Researchers attempt to introduce non-Euclidean latent spaces to explore more flexible priors. For example, Nickel and Kiela [64] mapped data into an n -dimensional Poincaré space to capture hierarchical representations. Liu and Zhu [65] used Riemannian manifolds as the latent space for inference models. However, these methods still require prior information and overlook the inherent spherical nature of some data. The vMF distribution is another competitive choice, which places the latent representation on the surface of a unit hypersphere to allow for uninformed priors [66], [67]. For example, Guu et al. [68] set the latent space of VAE as a vMF distribution and applied it to sentence editing tasks. Davidson et al. [69] replaced the Gaussian distribution with the vMF distribution and extended the reparameterization trick for rejecting sampling. Guan and Smith [70], Xu and Durrett [71], and Serban et al. [72] also reported superiorities of spherical latent spaces, which inspired us.

III. PRELIMINARIES

Here, we introduce the concepts of GAT, VAE, and vMF distribution and analyze the limitations of Gaussian distribution.

A. Graph Attention Networks

Given a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ with d nodes, where $\mathcal{V} = \{v_1, v_2, \dots, v_d\}$ is the node set. $\mathcal{E}$ is the edge set. Thus, the attention score [48] is calculated as

$$\alpha_{i,j} = \frac{\exp(\delta(\omega^\top \cdot (v_i \oplus v_j)))}{\sum_{k \in \mathcal{N}_i} \exp(\delta(\omega^\top \cdot (v_i \oplus v_k)))} \quad (1)$$

where $\delta(\cdot)$ is the *LeakyReLU* activation function; ω is the learnable weight matrix; $\oplus$ is the concatenation operation; $\mathcal{N}_i$ is the set of neighboring nodes of node i ; The output representation of the i th node is

$$h_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{i,j} v_j \right) \quad (2)$$

where $\sigma(\cdot)$ is the *Sigmoid* activation function.

B. Variational Autoencoder

The VAE is based on variational Bayesian inference [11]. Let $x \in \mathcal{X}$ represent the observed vector and $z \in \mathcal{Z}$ represent the continuous latent variable. The prior $p(z)$ conforms to a Gaussian distribution. Moreover, we use a neural network to parameterize the conditional distribution $p_\theta(x|z)$ for modeling the data distribution. We will train VAE by maximizing

marginal log-likelihood $\log p_\theta(x)$. However, directly optimizing $\log p_\theta(x)$ is complex. Therefore, a posterior distribution $p_\theta(z|x)$ is introduced. Specifically, we use a parameterized variational posterior $q_\phi(z|x)$ to approximate the true posterior. Based on this, marginal log-likelihood can be written as

$$\begin{aligned}\log p_\theta(x) &= \int_{\mathcal{Z}} q_\phi(z|x) \log p_\theta(x) dz \\ &= \int_{\mathcal{Z}} q_\phi(z|x) \log \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \cdot \frac{q_\phi(z|x)}{p_\theta(z|x)} \right] dz \\ &= \int_{\mathcal{Z}} q_\phi(z|x) \log \frac{p_\theta(x, z)}{q_\phi(z|x)} dz \\ &\quad + \int_{\mathcal{Z}} q_\phi(z|x) \log \frac{q_\phi(z|x)}{p_\theta(z|x)} dz \\ &= \mathcal{L}(\theta, \phi; x) + \text{KL}(q_\phi(z|x) || p_\theta(z|x)).\end{aligned}\quad (3)$$

From this, the first term is called the evidence lower bound (ELBO). The second term is the KL divergence of the variational posterior $q_\phi(z|x)$ and the true posterior $p_\theta(z|x)$. Hence, we often use $\mathcal{L}(\theta, \phi; x)$ as the lower bound of marginal log-likelihood [73].

Thus, we can expand $\mathcal{L}(\theta, \phi; x)$

$$\begin{aligned}\mathcal{L}(\theta, \phi; x) &= \int_{\mathcal{Z}} q_\phi(z|x) \log \frac{p_\theta(x, z)}{q_\phi(z|x)} dz \\ &= \int_{\mathcal{Z}} q_\phi(z|x) \log \frac{p(z)}{q_\phi(z|x)} dz \\ &\quad + \int_{\mathcal{Z}} q_\phi(z|x) \log p_\theta(x|z) dz \\ &= -\text{KL}(q_\phi(z|x) || p(z)) + \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].\end{aligned}\quad (4)$$

Intuitively, the first term encourages the variational posterior $q_\phi(z|x)$ to match the prior $p(z)$. The second term is the reconstruction loss. Since the Gaussian is a location-scale family of distributions, we can use reparameterization tricks to approximate ELBO.

C. Limitations of Gaussian Distribution

ELBO contains two gradient terms. One is the KL divergence term. Another one is the data likelihood (reconstruction loss). When the parameterized neural networks are RNNs, the KL divergence term regularizes the variational posterior $q_\phi(z|x)$ toward the prior $p(z)$ [71], [74]. Because RNNs are universal approximators that can express arbitrary distribution and do not rely on latent variables. According to [63], the variational posterior $q_\phi(z|x)$ carries very little information, which forces the model to ignore latent variables (the model considers latent variables as noise sources). Based on this, the model may fall into a local optimum, where $q_\phi(z|x) = p(z)$, which makes the KL divergence term in (4) equal to zero. In other words, this issue is KL collapse.

Zhao et al. [74] provided theoretical explanations for the above issue from the perspective of information preference. We rewrite (4) as [11]

$$\begin{aligned}\mathcal{L}(\theta, \phi; x) &= -\text{KL}(p_{\text{data}}(x) || p_\theta(x)) \\ &\quad - \mathbb{E}_{p_{\text{data}}(x)}[\text{KL}(q_\phi(z|x) || p_\theta(z|x))] \leq 0\end{aligned}\quad (5)$$

where p_{data} is the real data distribution. We assume that the distribution family for $p_\theta(x|z)$ is flexible [74]. So, there exists p^* in the distribution family that makes $\text{KL}(p_{\text{data}}(x) || p^*(x)) = 0$. Based on this, we make $p_\theta(z|x) = p^*$ and can obtain that the first term in (5) equals zero. When the second term in (5) is also zero, we obtain the optimal ELBO. In this case, we can get $p_\theta(z|x) = q_\phi(z|x) = p(z)$. The above situation demonstrates that the latent variable z provides no information.

D. vMF Distribution

The vMF distribution [69] is placed on the $(n-1)$ dimensional sphere $\mathbb{S}^{n-1}$. It is parameterized by a direction vector μ and a concentration parameter $\kappa \geq 0$, where $\|\mu\| = 1$. If $\kappa = 0$, the vMF is a Uniform distribution. The Probability Density Function of the vMF distribution for the unit vector z is defined as

$$q(z; \mu, \kappa) = C_n(\kappa) \exp(\kappa \mu^\top z) \quad (6)$$

$$C_n(\kappa) = \frac{\kappa^{n/2-1}}{(2\pi)^{n/2} \mathcal{I}_{n/2-1}(\kappa)} \quad (7)$$

where $\mathcal{I}$ represents the modified Bessel function.

IV. PROPOSED METHOD

A. Problem Statement

Given a multivariate time series $\mathcal{X} = \{x_1, x_2, \dots, x_T\}$, where $x_t \in \mathbb{R}^m$ is the measurement value of m sensors at timestamp t . We use a sliding window with a step size of τ to divide it into the sub-sequence set $\mathcal{S} = \{s_1, s_{1+\tau}, \dots\}$, where $s_t = \{x_t, x_{t+1}, \dots, x_{t+l-1}\}$ and l is the window size. This article aims to reconstruct sub-sequences without accessing label information, and it takes the deviation of reconstruction and actual values as the anomaly score.

B. Overview

Next, we describe VGATSL in detail. As indicated in Fig. 3, the model comprises three sub-modules: graph feature extractor, VAE with vMF distribution, and self-supervised learning module. Further, the workflow of this model is as follows.

- 1) Add perturbations to the original sub-sequence s ; Generate the abnormal instance $\tilde{s}$.
- 2) Graph feature extractor captures spatial-temporal information and obtains output representations h and $\tilde{h}$.
- 3) Establish a discriminator f_c to classify output representations h and $\tilde{h}$. It can help the model learn more explicit normality boundaries.
- 4) Encoder f_e maps the output representation h to a latent space $\mathcal{H} \in \mathbb{R}^n$. Meanwhile, the latent variable $z \in \mathbb{R}^n$ conforms to the vMF distribution.
- 5) Decoder f_d reconstructs the sub-sequence based on the latent variable z .

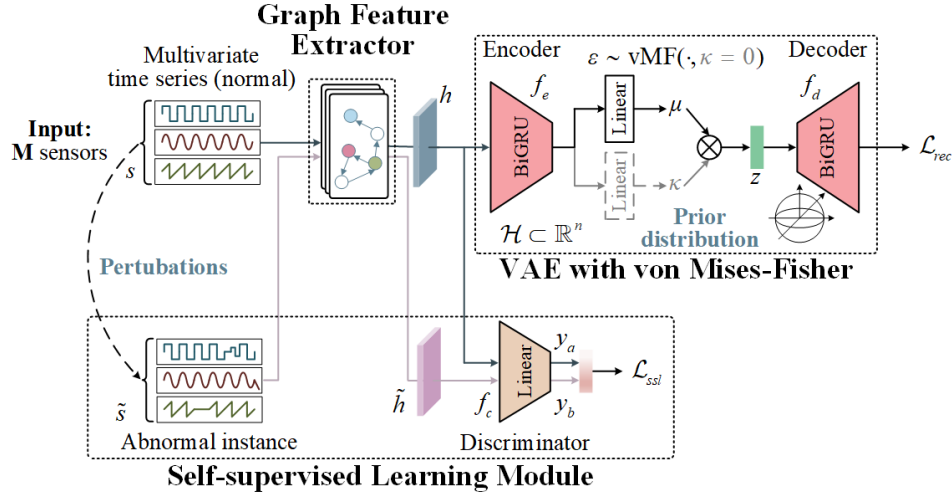


Fig. 3. Structure of VGATSL. It consists of three sub-modules: graph Feature Extractor, VAE with vMF distribution, and self-supervised learning module.

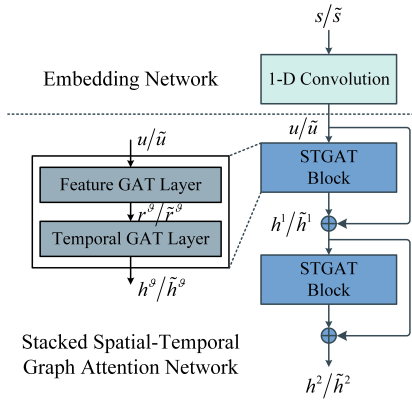


Fig. 4. Framework of graph feature extractor.

C. Graph Feature Extractor

This extractor aims to encode the spatial-temporal information of sub-sequences. It includes an embedding network and stacked STGAT networks [20], as indicated in Fig. 4.

The Embedding Network extracts the unique characteristics of each sensor [19]. Specifically, we use a 1-D CNN to transform the sub-sequence into a feature information matrix

$$u = \sigma(\text{Conv1D}(s, \mathbf{W}) + \mathbf{b}) \quad (8)$$

where $\text{Conv1D}(\cdot)$ represent 1-D convolution operation; $\mathbf{W}$ is the weight; $\mathbf{b}$ is the bias. This operation enables the subsequent stacked STGAT network to understand and process them.

The stacked STGAT Networks use attention mechanisms to capture the sub-sequences causal relationships and temporal dependencies. We first construct graph structures in feature and temporal dimensions.

- 1) *Feature-Oriented Graph Structure*: We treat the sub-sequence s_t as a complete graph $\mathcal{G}^{\text{feat}} = (\mathcal{V}^{\text{feat}}, \mathcal{E}^{\text{feat}})$, where $\mathcal{V}^{\text{feat}} = \{v_1, v_2, \dots, v_m\}$ is the node set, and the sequence $s_t^{(i)}$ at sensor i represents node v_i . $\mathcal{E}^{\text{feat}}$ is the edge set. A correlation relationship between any nodes i and j must exist. There may be implicit dependency

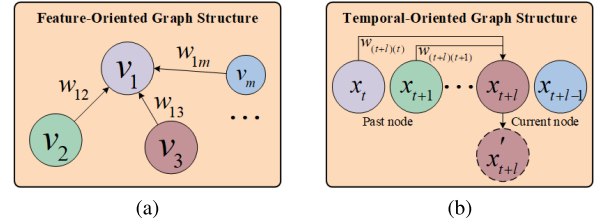


Fig. 5. Graph Structures. (a) Feature-oriented graph structure. (b) Temporal-oriented graph structure.

relationships between sensors. So, we set the adjacent matrix $A_{ij}^{\text{feat}} = 1$, where weight w_{ij} , $i, j \in [1, m]$. Indeed, the dependency patterns between sensors are symmetrical. Thus, the adjacent matrix A^{feat} should be diagonal. In other words, we believe the nodes have maximum connectivity, and each edge has a weight. Fig. 5(a) shows the structure.

- 2) *Temporal-Oriented Graph Structure*: Meanwhile, we treat all timestamps within a sliding window as a complete graph $\mathcal{G}^{\text{tmp}} = (\mathcal{V}^{\text{tmp}}, \mathcal{E}^{\text{tmp}})$, where $\mathcal{V}^{\text{tmp}}$ is the node set. Each node is a feature vector x_{t+c} at the timestamp $t + c$, and $0 \leq c \leq l - 1$ represents the time interval. $\mathcal{E}^{\text{tmp}}$ is an edge set that reflects the connection relationship between different timestamps. Fig. 5(b) shows the structure. We believe there is a connection between the past and current nodes [20]. It is reasonable. A sensor signal is a collection of data points arranged in time order. Random variables change over time and have continuity. Thus, we set the adjacent matrix $A_{ab}^{\text{tmp}} = 1$, where $a \geq b$. In contrast, $A_{ab}^{\text{tmp}} = 0$, where $a < b$, and $a, b \in [t, t + l - 1]$.

Then, we use the STGAT block to process feature-oriented and temporal-oriented graph-structured data, as indicated in Fig. 4. Indeed, the STGAT block contains a feature GAT layer and a temporal GAT layer. Specifically, we input the matrix u into the feature GAT layer to learn the spatial correlation, according to (1) and (2). The output representation r^ϑ can be obtained, where ϑ represents the number of STGAT blocks.

Meanwhile, the same operation captures temporal correlation in different timestamps. So, we can obtain the final output representation h^θ .

D. VAE With vMF

Based on the above discussion, we adopt the VAE model [11] to achieve the reconstruction task and use BiGRUs as the encoder f_e and decoder f_d . Compared with LSTM-based models, BiGRUs have fewer parameters, faster convergence speed, and superior performance [75]. Additionally, we choose the vMF distribution as the prior and variational posterior to avoid KL collapse [63], [74].

Specifically, the encoder f_e encodes the output representation h and obtains the latent variable $z \in \mathbb{R}^n$. Here, we assume that the prior $p(z)$ is a Uniform distribution on the hypersphere $\text{vMF}(\cdot, \kappa = 0)$. It is represented as one divided by the surface area of a $(n - 1)$ dimensional sphere $1/[2(\pi^{n/2})/\Gamma(n/2)]$, where $\Gamma(\cdot)$ is the *Gamma* function [69]. Meanwhile, we use the variational posterior $q_\phi(z|x) = \text{vMF}(z; \mu, \kappa)$ to approximate the true posterior $p_\theta(z|x)$, where the mean parameter μ is the output of the encoding neural network and the concentration parameter κ is a constant. According to (6) and (7), we can rewrite the KL divergence term in (4)

$$\begin{aligned} & \text{KL}(\text{vMF}(z; \mu, \kappa) || \text{vMF}(\cdot, 0)) \\ &= \text{KL}(q_\phi(z|x) || p(z)) \\ &= \int_{\mathcal{Z}} q_\phi(z|\mu, \kappa) \log \frac{q_\phi(z|\mu, \kappa)}{p(z)} dz \\ &= \int_{\mathcal{Z}} q_\phi(z|\mu, \kappa) (\log C_n(\kappa) + \kappa \mu^T z - \log p(z)) dz \\ &= \log C_n(\kappa) + \kappa \mu \mathbb{E}_q[z] + \log \left[\frac{2(\pi^{n/2})}{\Gamma(n/2)} \right] \\ &= \left(\frac{n}{2} - 1 \right) \log \kappa - \frac{n}{2} \log(2\pi) - \log \mathcal{I}_{\frac{n}{2}-1}(\kappa) \\ &\quad + \kappa \frac{\mathcal{I}_{\frac{n}{2}}(\kappa)}{\mathcal{I}_{\frac{n}{2}-1}(\kappa)} + \frac{n}{2} \log \pi + \log 2 - \log \Gamma\left(\frac{n}{2}\right). \end{aligned} \quad (9)$$

It is evident that the modified KL divergence relies solely on the concentration κ , and not on the mean μ . KL divergence is a function of hyperparameters n and κ , which can be precisely controlled [68]. Specifically, we consider κ as a constant. Thus, the KL divergence term is also a constant, and the KL collapse will not occur; there is $q_\phi(z|x) \neq p(z)$. At this point, we believe the latent variable z provides rich information. However, when using Gaussian VAE, the value of KL divergence is constantly changing. It exhibits a concentrated probability mass around the origin. In other words, the Gaussian VAE habit pulls the posterior toward the prior [71]. In this case, the latent variable z does not provide any information, which makes optimization difficult [76].

Next, we introduce the details of sampling latent variable z from the vMF distribution. We follow Wood's rejection sampling scheme [77], as illustrated in Algorithm 1. It is an improved Ulrich's procedure [78]. Specifically, we sample ε from the distribution $\text{vMF}(z|\mathbf{e}_1, \kappa)$ with modal vector $\mathbf{e}_1 = (1, 0, \dots, 0)$, where ε is a unit vector tangent to the hypersphere. Second, we use an acceptance-rejection sampling

scheme [77] to sample $\varpi \in [-1, 1]$ from the univariate density $g(\varpi|\kappa, n)$. According to [69], we can expand the univariate density $g(\varpi|\kappa, n)$ as follows:

$$g(\varpi|\kappa, n) = \frac{2(\pi^{\frac{n}{2}})}{\Gamma(\frac{n}{2})} \frac{\exp(\kappa \varpi) (1 - \varpi^2)^{\frac{1}{2}(n-3)}}{B(\frac{1}{2}, \frac{1}{2}(n-1))} \quad (10)$$

where $B(\cdot)$ is the *Beta* function. Thus, we can calculate the latent variable $z' \leftarrow (\varpi; ((1 - \varpi^2)^{1/2})\varepsilon^\top)^\top$ under modal $\mathbf{e}_1$. Finally, we use orthogonal transform $U(\mu)$ to obtain a latent variable z that conforms to $q(z|\mu, \kappa)$. Specifically, we use *Householder* reflection to implement this operation, as shown in Algorithm 2.

Algorithm 1 Sampling Process

Require: Mean μ ; Concentration κ ;

- 1: Sample $\varepsilon \sim \text{vMF}(z|\mathbf{e}_1, \kappa)$;
 - 2: Sample $\varpi \sim g(\varpi|\kappa, n) \propto \exp(\kappa \varpi) (1 - \varpi^2)^{(n-3)/2}$;
 - 3: $z' \leftarrow (\varpi; (\sqrt{1 - \varpi^2})\varepsilon^\top)^\top$;
 - 4: $U \leftarrow \text{Householder}(\mathbf{e}_1, \mu)$;
- return** Uz' ;
-

Algorithm 2 Householder Transform

Require: Mean μ ; Modal vector $\mathbf{e}_1$;

- 1: $v' \sim \mathbf{e}_1 - \mu$;
 - 2: $v \leftarrow \frac{v'}{\|v'\|}$;
 - 3: $U \leftarrow \mathbb{I} - 2vv^\top$;
- return** U ;
-

On this basis, we utilize decoder f_d to reconstruct the sub-sequence. Its input is latent variable z . The loss function is

$$\begin{aligned} \mathcal{L}_{\text{rec}} &= \mathbb{E}_{q_\phi(z|s)} [\log p_\theta(s|z)] \\ &\quad - \text{KL}(\text{vMF}(\mu, \kappa) || \text{vMF}(\cdot, \kappa = 0)) \end{aligned} \quad (11)$$

where the first term is reconstruction loss, and the second term is the KL divergence based on vMF distribution.

E. Self-Supervised Learning

This module uses pretext tasks to provide information about anomalies. Actually, we add tailored perturbations to normal sub-sequences to simulate abnormal instances. These instances help to learn more explicit decision boundaries. Specifically, contextual and collection transformations are adopted [60].

- 1) *Contextual Transformation*: Randomly select dimensions of the last observation of the input sub-sequence; Calculate the mean of the previous values; Add offset to the mean and obtain the contextual outlier; Replace the data values of the selected dimensions with contextual outliers. The offset is -0.5 or $+0.5$.
- 2) *Collection Transformation*: Randomly select a group of dimensions; Randomly select a segment of the input sub-sequence (its length not exceeding 0.2 l); Take the last observation as the endpoint of the segment; Use the collection outliers to replace the values of the segment in the selected dimension.

TABLE I
INFORMATION OF DATASETS

Dataset	Dims	Train	Test	Anomalies
SWaT	51	47515	44986	11.97%
WADI	127	118795	17275	5.99%
WTD	10	24133	28657	43.70%

To learn intrinsic representations from these transformations, we use a discriminator f_c to classify the output representations h and $\tilde{h}$, as shown in Fig. 3. It helps determine whether the input sub-sequence is from the normal or abnormal data. So, this process can be formulated as

$$\mathcal{L}_{\text{ssl}} = -[y_a \log(f_c(h)) + y_b \log(f_c(\tilde{h}))] \quad (12)$$

where $y_a = 1$ and $y_b = 0$ are pseudo labels. Please note that abnormal instances share the same pseudo labels.

F. Training and Inference

1) *Training*: In summary, we combine reconstruction and self-supervised losses to train the model. The final loss function is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{ssl}} \quad (13)$$

where λ is a tradeoff coefficient.

2) *Inference*: The VAE with vMF distribution attempts to reconstruct the input sub-sequence. We use unlabeled historical normal data to train the model. Thus, based on the reconstruction error value, we can judge whether the test instance is normal or abnormal.

First of all, we calculate the abnormal score as

$$\text{Score} = \frac{1}{m} \|x_t - \hat{x}_t\|_2 \quad (14)$$

where $\hat{x}_t$ is the predicted value. Second, consistent with [20], we set a decision threshold based on expert knowledge. If the abnormal score exceeds the threshold, the input is abnormal and vice versa.

V. EXPERIMENTAL EVALUATION

A. Datasets and Metrics

1) *Datasets*: We evaluate and compare VGATSL on three public and real-world datasets. As listed in Table I, we provide helpful information such as dataset name and size, sample dimension, and anomaly rate.

- 1) *Secure Water Treatment Dataset (SWaT)* [16] comes from a water treatment testbed with 51 sensors. The system has been running for 11 days. Further, the system suffered 41 cyber and physical attacks in the last 4 days. These attacks have different intentions and durations. Please note that the collection is continuous. Therefore, we label the overall sequential data based on each timestamp's normal and abnormal behavior. In the experiment, we use 47515 normal data collected in the first 7 days for training and 44986 data collected in the last 4 days for testing. The anomaly rate in the test

set is 11.97%. We downsample the SWaT dataset every 10 seconds to speed up the training.

- 2) *Water Distribution Dataset (WADI)* [79] is a simulated large-scale urban water distribution system with more realistic water treatment and distribution networks. The system has been running for 16 days. In the first 14 days, we collected 118 795 normal data for training—the last 2 days in attack scenarios. Thus, we collected 17 275 data for testing, of which 5.99% were abnormal. The testbed consists of 127 sensors and actuators. We also downsample the WADI dataset.
- 3) *Wind Turbine Dataset (WTD)* [20] is collected by ten sensors. The collection process lasts 1–2 years, with a sampling interval of 10 min. Considering on-the-spot operation and maintenance records from wind farm operators, we labeled the data deviating from normal mode. The proportion of abnormal data in the test set is 43.70%.

Based on this, the data format is measured values with timestamps, known as a multivariate time series. We use all sensor records as input features for the model. Specifically, we set sliding step $\tau = 1$ and the sliding window size $l = 5$ under SWaT, $l = 5$ under WADI, and $l = 60$ under WTD. Totally, we obtain 47 511 training samples and 44 982 test samples under SWaT (sample format is sub-sequence, with feature size of 51 and length of 5), 118 791 training samples and 17 271 test samples under WADI (feature size of 127 and length of 5), and 24 074 training samples and 28 598 test samples under WTD (feature size of 10 and length of 60). Please note that once a timestamp in a sub-sequence exhibits abnormal behavior, we will label the entire sub-sequence as an abnormal sample. Furthermore, we apply the same data preprocessing to all methods, i.e., using Minimum-and-Maximum normalization to scale the SWaT, WADI, and WTD datasets. We randomly divide 20% of the training data as the validation set.

2) *Metrics*: Consistent with [5], [6], and [19], we select Precision (Prec), Recall (Rec), and F1-Score (F1) as evaluation indicators.

B. Comparison Methods

VGATSL is compared with different popular and typical methods.

- 1) *PCA* [80], which is used to find low-dimensional projections to capture most of the variance in the data.
- 2) *iForest* [27], which isolates each data point to a termination node. The path length from the termination node to the root node is an anomaly measure.
- 3) *LSTM-AE* [10], which is a reconstruction-based method. We set the encoder and decoder as a double-layer LSTM.
- 4) *USAD* [81], which is a typical unsupervised anomaly detection method that uses an adversarial training scheme based on encoder–decoder architecture to learn reconstruction errors.
- 5) *MAD-GAN* [3], which is a GAN-based method that uses LSTM-RNN to capture temporal correlations.

TABLE II
COMPARISON OF DIFFERENT METHOD

Method	SWaT			WADI			WTD		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
PCA	0.2915	0.2865	0.2890	0.2228	0.2317	0.2272	0.3603	0.3623	0.3613
iForest	0.5178	0.7053	0.5972	0.1868	0.5868	0.2834	0.4154	0.3081	0.3538
LSTM-AE	0.9701	0.5730	0.7205	0.5256	0.2126	0.3027	0.8810	0.7792	0.8270
USAD	0.9945	0.5737	0.7276	0.9957	0.1516	0.2631	0.8325	0.8663	0.8491
MAD-GAN	0.9874	0.6338	0.7720	0.9751	0.1294	0.2285	0.4627	0.7719	0.5786
GDN	0.9928	0.6096	0.7554	0.7724	0.3082	0.4406	0.8288	0.7451	0.7847
MTAD-GAT	0.9673	0.6094	0.7477	0.5119	0.2282	0.3157	0.8311	0.8380	0.8345
STGAT-MAD	0.9772	0.6497	0.7805	0.9285	0.2862	0.4375	0.8483	0.8655	0.8568
VGATSL	0.9511	0.6656	0.7831	0.7759	0.3185	0.4516	0.8807	0.8903	0.8855

- 6) *GDN* [19], which learns the topological relationships between sensors and utilizes attention mechanisms to predict the future values of nodes.
- 7) *MTAD-GAT* [6], which includes reconstruction and prediction-based models, treating all timestamps and variables in a multivariate time series as two complete graphs to encode temporal and spatial correlations.
- 8) *STGAT-MAD* [20], which uses stacked STGAT networks to capture the temporal and feature correlations.

The network structure of VGATSL is as follows: 1) the number of STGAT blocks is two; 2) encoder or decoder is BiGRUs with 150 neurons; 3) the dimension size of z is 15; and 4) the discriminator is a double-layer, fully connected network.

We train this model on an NVIDIA GeForce RTX 4090 GPU. The Adam optimizer [82] is also selected, and the initial learning rate is 0.001. We set the hyperparameter $\lambda = 0.1$, and $\kappa = 25$.

C. Results and Analysis

Table II reports the results. VGATSL achieved the best results on all datasets. Compared with other methods, VGATSL improves the *F1*-Score by 0.26% on SWaT dataset, 1.10% on WADI dataset, and 2.86% on WTD dataset. Overall, as the size of the dataset increases, there is less performance improvement. The main reasons are as follows.

- 1) Inaccurate label information.
- 2) Data contains more noise.
- 3) Differences in data distribution.

Because of the limitations of feature extraction techniques, traditional methods perform terribly on three datasets. In deep learning methods, LSTM-AE and USAD are susceptible to noise interference, which makes identifying normality boundaries in large-scale datasets challenging. MAD-GAN performed well on the SWaT dataset. However, it performs terribly on WADI and WTD datasets. Because the generative ability of MAD-GAN is limited, it cannot capture all patterns in complex data distributions. Graph learning methods, such as GDN, MTAD-GAT, and STGAT-MAD, can extract spatial information, which helps improve detection performance. However, these methods make it difficult to model the stochastic interrelationships during reconstruction.

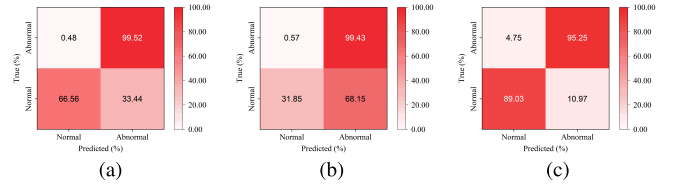


Fig. 6. Confusion matrices on three datasets. (a) SWaT. (b) WADI. (c) WTD.

The competitiveness of VGATSL can be attributed to these aspects. First, we use advanced network structures to model multivariate time series, which compensate for the shortcomings of existing methods in spatial-temporal information loss. Second, different signal transformations collaborate to generate anomaly instances, which helps the model form more explicit decision boundaries. Third, compared with the Gaussian distribution, the vMF distribution learns stochastic interrelationships, which helps to eliminate the KL collapse of the Encode-Decode.

Next, we plot VGATSL's confusion matrices. As indicated in Fig. 6, VGATSL can effectively identify the abnormal class. For SWaT dataset, the actual negative rate is 99.52%. For WADI dataset, the actual negative rate is 99.43%. For WTD dataset, the actual negative rate is 95.25%. The misclassification proportion is higher for normal data than for abnormal data. This can be attributed to:

- 1) We do not use the "Point Adjust" scheme in this experiment.
- 2) We do not use the "Reduction," which can confuse the decision boundaries.

D. Ablation Study

Meanwhile, ablation studies demonstrate the necessity of different components. We set five variants.

- 1) Disable graph feature extractor (abbreviated as w/o GFE).
- 2) Disable vMF distribution (replace with Gaussian distribution, and abbreviated as w/o vMF, G-VAE).
- 3) Disable vMF distribution (replace with AE, abbreviated as w/o vMF, AE).
- 4) Disable self-supervised learning module (abbreviated as w/o SSL).

TABLE III
ABLATION RESULTS

Method	WTD		
	Prec	Rec	F1
VGATSL	0.8807	0.8903	0.8855
w/o GFE	0.8210	0.9182	0.8669
w/o vMF (G-VAE)	0.8250	0.8770	0.8502
w/o vMF (AE)	0.8321	0.8857	0.8581
w/o SSL	0.8340	0.8497	0.8418
w/o BiGRUs	0.7721	0.8092	0.7901

5) Disable BiGRUs (replace with BiLSTM, and abbreviated as w/o BiGRUs).

Table III shows the results. After removing the graph feature extractor, we observe that the $F1$ -Score decreases from 0.8855 to 0.8669. The feature extractor can encode spatiotemporal information, which helps the model capture the causal relationships and temporal dependencies. We also find that w/o SSL reduces performance. Indeed, this module helps the model learn more explicit normality boundaries.

The vMF distribution allows for uninformed priors, thus avoiding the KL collapse. After replacing the latent space with Gaussian distribution, $F1$ -Score decreased by 3.53%; After replacing the latent space with AE, $F1$ -Score decreased by 2.74%. It is because Gaussian VAE pulls the posterior toward the prior. Moreover, BiGRUs are function approximators that can fit any probability distribution and do not rely on latent variables. AE's performance is slightly better than G-VAE.

If the BiLSTM backbone network was used, $F1$ -Score decreased by 9.54% compared to the proposed method. BiGRUs have fewer parameters, faster convergence speed, and a superior ability to capture time series features.

E. Parameter Sensitivity Analysis

To better illustrate the impact of different parameters, we set three configurations: sliding window length l , tradeoff parameter λ , and dimensions of latent variables n .

This section provides a detailed performance comparison of the WTD dataset. Fig. 7(a) shows the comparison results of different sliding window lengths $l \in [5, 30, 60, 100]$. Here, we set the tradeoff parameter to 0.1 and the dimensions of latent variables to 15. The results indicate that the performance gradually improves as the sliding window length increases. When $l = 100$, $F1$ -Score suddenly drops to 0.7797. This is because a larger sliding window may not only provide more temporal information to the model but also increase the difficulty of encoding temporal representations.

In Fig. 7(b), we compare the test results with different tradeoff parameters $\lambda \in [0.1, 0.4, 0.7, 1.0]$. It can be observed that as the tradeoff parameter λ gradually increases, the overall performance shows a downward trend. When $\lambda = 0.7$, the model achieves the lowest $F1$ -Score. Self-supervised learning brings more raw anomaly information to the model. However, during training, a larger proportion of self-supervised learning losses can affect the performance of other components and mislead the decision boundaries.

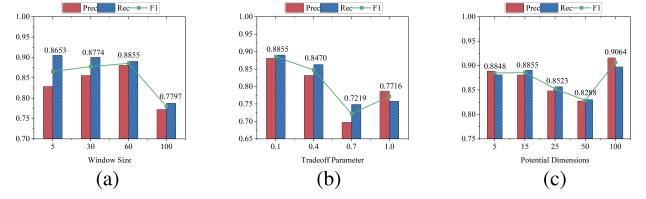


Fig. 7. Parameter sensitivity analysis on wind turbine Dataset. (a) SWaT. (b) WADI. (c) WTD.

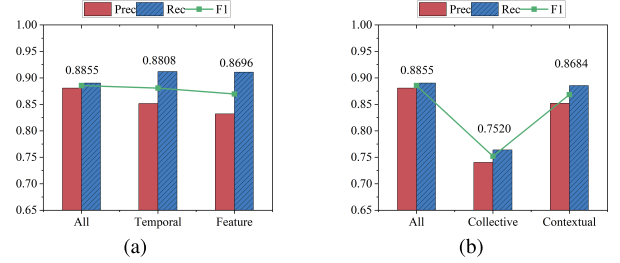


Fig. 8. Detailed analysis of the proposed method. (a) Graph attention network. (b) Self-supervised learning.

Furthermore, we evaluated the performance of dimensions for different latent variables $n \in [5, 15, 25, 50, 100]$. Here, we set the sliding window length to 60 and the tradeoff parameter to 0.1. As indicated in Fig. 7(c), VGATSL is robust to the latent dimensions, and the performance difference is insignificant under different parameter settings. We find that when $n = 50$, $F1$ -Score decreases to 0.8288; When $n = 100$, $F1$ -Score reaches 0.9064.

F. Detailed Analysis

Previous experiments have demonstrated the importance of the graph feature extractor. In Fig. 8(a), we analyze the performance of different attention layers (i.e., feature and temporal graph attention layer). The experiment is conducted on the WTD dataset. When only using the feature or temporal graph attention layer, the $F1$ -Score significantly decreases, with an accuracy decrease of over 3%. This indicates that the relationship between the feature and the temporal graph attention layers is complementary, playing a positive role in extracting spatial and temporal information. We find that under the same conditions, temporal-oriented graph attention networks are superior to feature-oriented graph attention networks. This is because industrial data usually has temporal characteristics such as dynamism and nonstationarity, which contain the evolution laws and state information of industrial systems and are more prominent than spatial characteristics such as coupling in topological relationships.

Fig. 8(b) shows the comparative performance analysis of different signal transformations. It can be observed that the signal transformation of “Contextual” is competitive. When the model only learns the enhanced data generated by the “Collection” signal transformation, its performance will significantly decrease. We believe the anomaly examples generated by the transformation of “Collection” are relatively rare in realistic datasets. When the model is trained, its decision boundaries will be blurred if all the enhanced data are such examples.

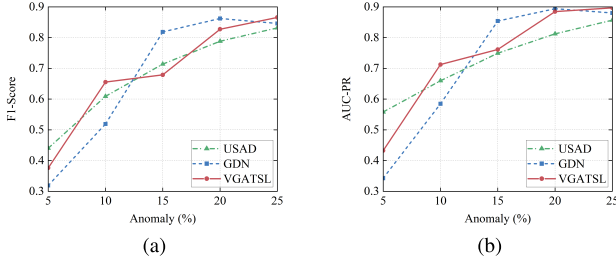


Fig. 9. Performance comparison under different percentage of anomaly. (a) $F1$ -Score. (b) AUC-PR.

However, when the model jointly learns the enhanced data generated by the transformation of two types of signals, it will achieve complementary performance and achieve the best detection results.

G. Percentage of Anomaly

Next, we set the anomaly ratios on the test set to 5%, 10%, 15%, 20%, and 25% to simulate possible anomalies that may occur in the real world. We still use the WTD dataset as an example. In particular, we chose USAD and GDN as a comparison.

Fig. 9(a) shows the $F1$ -Score under different anomaly ratios. Meanwhile, we plot the corresponding AUC-PR values, as indicated in Fig. 9(b). Overall, the performance of VGATSL is superior to that of USAD and GDN. It is worth mentioning that when the ratio values are 15% and 20%, the performance of GDN reaches its best. This experiment demonstrates that VGATSL has good stability and can address the problem of imbalanced datasets in real industrial scenarios.

H. Case Studies

We provide cases to demonstrate the interpretability of VGATSL. Specifically, we use a downsampling technique to reduce the sequence length. As indicated in Fig. 10(a), we present test examples on the WTD dataset. The red area represents the abnormal data we have detected. On this basis, we calculate the average anomaly score, which helps to locate the faulty sensor. To save space, we only analyze the situation in region two. As shown in Fig. 10(b), darker colors indicate higher average anomaly scores; therefore, wind speed is considered the most likely sensor to fail. The anomaly is the cracking of the main bearing, and frequent changes in wind speed are the leading cause.

I. Convergence and Complexity Analysis

Fig. 11 shows the training loss of VGATSL. For the SWaT dataset, this method converges at the 50th epoch; For the WADI dataset, this method converges at the 80th epoch; For the WTD dataset, this method converges at the 500th epoch. In conclusion, VGATSL has convergence and stability and a fast convergence speed.

Next, we evaluate the inference time of various methods on different datasets. Here, we chose MAD-GAN, GDN, and STGAT-MAD as comparative methods. As indicated in Fig. 12, VGATSL is competitive regarding time efficiency.

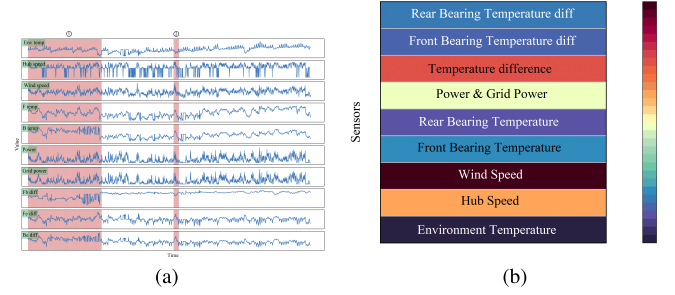


Fig. 10. Case analysis on wind turbine dataset. (a) Multivariate time series. (b) Anomaly score.

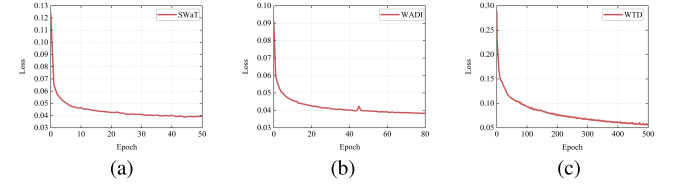


Fig. 11. Convergence of proposed method. (a) SWaT. (b) WADI. (c) WTD.

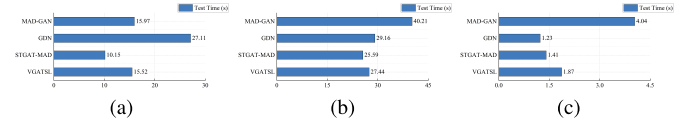


Fig. 12. Inference time of proposed method. (a) SWaT. (b) WADI. (c) WTD.

Method	Parameters	Model size
MAD-GAN	0.56M	2.14MB
GDN	0.01M	0.04MB
STGAT-MAD	0.79M	3.01MB
VGATSL	0.48M	1.84MB

Table IV lists the parameter sizes of different models to compare their corresponding spatial efficiency. GDN has the lowest number of parameters and model size. Additionally, VGATSL achieves the second-best spatial efficiency. Compared with MAD-GAN, its model size has decreased by 0.3 MB; Compared with STGAT-MAD, its model size decreased by 1.17 MB. Therefore, the proposed method also has flexibility in practical applications.

VI. LIMITATIONS AND FUTURE WORK

Although the proposed method reduces computational costs (memory consumption, FLOP, parameter count), it takes 15.52 s (0.345 ms/sample) to infer SWaT, 27.44 s (1.589 ms/sample) to infer WADI, and 1.87 s (0.065 ms/sample) to infer WTD on NVIDIA GeForce RTX 4090 GPU. We consider this to be due to multiple spatiotemporal attention blocks in the network. In addition, solving the vMF distribution is relatively cumbersome, and using reparameterization techniques can reduce application efficiency and feasibility.

In the future, we will research: 1) we attempt to incorporate prior knowledge into the proposed method to learn intrinsic representations from mixed information sources and

achieve fault localization, monitoring, and other activities in more complex industrial systems and 2) we will develop more efficient and robust models and conduct comprehensive theoretical analysis.

VII. CONCLUSION

This article proposes an unsupervised anomaly detection method called variational graph attention networks with self-supervised learning. We have built a graph feature extractor based on stacked STGAT networks to learn feature-oriented and temporal-oriented topological structures. In particular, the relationship between the feature and the temporal graph attention layers is complementary, playing a positive role in extracting spatial and sequential information. Additionally, the vMF distribution is selected as the latent space, which places the latent representation on the surface of the unit hypersphere to allow for uninformed priors, thus preventing the model from falling into the local optimum. We also find that the performance of different signal transformations varies. The self-supervised learning module can simulate abnormal behaviors and help the model form more explicit decision boundaries. Extensive experiments demonstrate that VGATSL outperforms the baseline methods and can be applied to actual industrial scenarios.

REFERENCES

- [1] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2019, pp. 2828–2837.
- [2] Y. Zhang, J. Wang, Y. Chen, H. Yu, and T. Qin, "Adaptive memory networks with self-supervised learning for unsupervised anomaly detection," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12068–12080, Dec. 2023.
- [3] D. Li, D. Chen, B. Jin, L. Shi, J. Goh, and S.-K. Ng, "MAD-GAN: Multivariate anomaly detection for time series data with generative adversarial networks," in *Proc. Int. Conf. Artif. Neural Netw.* Cham, Switzerland: Springer, 2019, pp. 703–716.
- [4] B. Zhou, S. Liu, B. Hooi, X. Cheng, and J. Ye, "BeatGAN: Anomalous rhythm detection using adversarially generated time series," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 4433–4439.
- [5] W. Zhang, C. Zhang, and F. Tsung, "GRELEN: Multivariate time series anomaly detection from the perspective of graph relational learning," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, 2022, pp. 2390–2397.
- [6] H. Zhao et al., "Multivariate time-series anomaly detection via graph attention network," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2020, pp. 841–850.
- [7] K. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space Odyssey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2222–2232, Oct. 2016.
- [8] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager, "Temporal convolutional networks for action segmentation and detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 156–165.
- [9] I. J. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
- [10] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, "LSTM-based encoder-decoder for multi-sensor anomaly detection," 2016, *arXiv:1607.00148*.
- [11] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.
- [12] M. Gutoski, N. M. R. Aquino, M. Ribeiro, A. E. Lazzaretti, and H. S. Lopes, "Detection of video anomalies using convolutional autoencoders and one-class support vector machines," in *Proc. 13th Brazilian Congr. Comput. Intelligence*, Jan. 2018, pp. 1–12.
- [13] W. Mao, G. Wang, L. Kou, and X. Liang, "Deep domain-adversarial anomaly detection with one-class transfer learning," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 2, pp. 524–546, Feb. 2023.
- [14] Q. Qian, J. Luo, and Y. Qin, "Adaptive intermediate class-wise distribution alignment: A universal domain adaptation and generalization method for machine fault diagnosis," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 21, 2024, doi: [10.1109/TNNLS.2024.3376449](https://doi.org/10.1109/TNNLS.2024.3376449).
- [15] H. Li, W. Zheng, F. Tang, Y. Zhu, and J. Huang, "Few-shot time-series anomaly detection with unsupervised domain adaptation," *Inf. Sci.*, vol. 649, Nov. 2023, Art. no. 119610.
- [16] J. Goh, S. Adepu, M. Tan, and Z. S. Lee, "Anomaly detection in cyber physical systems using recurrent neural networks," in *Proc. IEEE 18th Int. Symp. High Assurance Syst. Eng. (HASE)*, Jan. 2017, pp. 140–145.
- [17] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Mar. 2020.
- [18] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3844–3852.
- [19] A. Deng and B. Hooi, "Graph neural network-based anomaly detection in multivariate time series," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, May 2021, pp. 4027–4035.
- [20] J. Zhan et al., "Stgat-mad: Spatial-temporal graph attention network for multivariate time series anomaly detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 3568–3572.
- [21] W. Chen, L. Tian, B. Chen, L. Dai, Z. Duan, and M. Zhou, "Deep variational graph convolutional recurrent network for multivariate time series anomaly detection," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 3621–3633.
- [22] L. Zhang et al., "Self-supervised variational graph autoencoder for system-level anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.
- [23] N. Chen, H. Tu, H. Zeng, and Y. Ou, "Anomaly detection for key performance indicators by fusing self-supervised spatio-temporal graph attention networks," *Knowl.-Based Syst.*, vol. 300, Sep. 2024, Art. no. 112167.
- [24] Q. Yu, L. Jibin, and L. Jiang, "An improved ARIMA-based traffic anomaly detection algorithm for wireless sensor networks," *Int. J. Distrib. Sensor Netw.*, vol. 12, no. 1, Jan. 2016, Art. no. 9653230.
- [25] J. Ma and S. Perkins, "Time-series novelty detection using one-class support vector machines," in *Proc. Int. Joint Conf. Neural Netw.*, vol. 3, 2003, pp. 1741–1745.
- [26] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.
- [27] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining*, Oct. 2008, pp. 413–422.
- [28] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTM and non-parametric dynamic thresholding," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2018, pp. 387–395.
- [29] M. Kravchik and A. Shabtai, "Detecting cyber attacks in industrial control systems using convolutional neural networks," in *Proc. Workshop Cyber-Phys. Syst. Secur. PrivaCy*, Jan. 2018, pp. 72–83.
- [30] G. Lai, W.-C. Chang, Y. Yang, and H. Liu, "Modeling long-and short-term temporal patterns with deep neural networks," in *Proc. Int. ACM Conf. Res. Devel. Inf. Retrieval (SIGIR)*, 2018, pp. 95–104.
- [31] X. Chen et al., "DAEMON: Unsupervised anomaly detection and interpretation for multivariate time series," in *Proc. IEEE 37th Int. Conf. Data Eng. (ICDE)*, Apr. 2021, pp. 2225–2230.
- [32] B. Zong et al., "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–19.
- [33] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.
- [34] Y. Zhang, Y. Chen, J. Wang, and Z. Pan, "Unsupervised deep anomaly detection for multi-sensor time-series signals," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 2, pp. 2118–2132, Feb. 2021.
- [35] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, Jul. 2009.
- [36] V. Vercauteren, W. Meert, and J. Davis, "Transfer learning for time series anomaly detection," in *Proc. IAL@ PKDD/ECML*, 2017, pp. 27–36.

- [37] V. Vincent, M. Wannes, and D. Jesse, "Transfer learning for anomaly detection through localized and unsupervised instance selection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 4, pp. 6054–6061.
- [38] T. Wen and R. Keyes, "Time series anomaly detection using convolutional neural networks and transfer learning," 2019, *arXiv:1905.13628*.
- [39] J. Shen, Y. Qu, W. Zhang, and Y. Yu, "Wasserstein distance guided representation learning for domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.
- [40] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*.
- [41] Q. Qian, Y. Wang, T. Zhang, and Y. Qin, "Maximum mean square discrepancy: A new discrepancy representation metric for mechanical fault transfer diagnosis," *Knowl.-Based Syst.*, vol. 276, Sep. 2023, Art. no. 110748.
- [42] Q. Qian, H. Pu, T. Tu, and Y. Qin, "Variance discrepancy representation: A vibration characteristic-guided distribution alignment metric for fault transfer diagnosis," *Mech. Syst. Signal Process.*, vol. 217, Aug. 2024, Art. no. 111544.
- [43] T. Cao, J. Zhu, and G. Pang, "Anomaly detection under distribution shift," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6488–6500.
- [44] Q. Wang, G. Pang, M. Salehi, W. Buntine, and C. Leckie, "Cross-domain graph anomaly detection via anomaly-aware contrastive alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 4, pp. 4676–4684.
- [45] F. Xia et al., "Graph learning: A survey," *IEEE Trans. Artif. Intell.*, vol. 2, no. 2, pp. 109–127, Apr. 2021.
- [46] J. Zhou et al., "Graph neural networks: A review of methods and applications," *AI Open*, vol. 1, pp. 57–81, Jan. 2020.
- [47] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
- [48] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
- [49] H. Hojjati, M. Sadeghi, and N. Armanfard, "Multivariate time-series anomaly detection with temporal self-supervision and graphs: Application to vehicle failure prediction," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*. Cham, Switzerland: Springer, 2023, pp. 242–259.
- [50] D. Grattarola, D. Zamboni, L. Livi, and C. Alippi, "Change detection in graph streams by learning graph embeddings on constant-curvature manifolds," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 6, pp. 1856–1869, Jun. 2020.
- [51] P. Qi, D. Li, and S.-K. Ng, "MAD-SGCN: Multivariate anomaly detection with self-learning graph convolutional networks," in *Proc. Int. Conf. Data Eng.*, 2022, pp. 1232–1244.
- [52] J. Wang, S. Shao, Y. Bai, J. Deng, and Y. Lin, "Multiscale wavelet graph AutoEncoder for multivariate time-series anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.
- [53] T. K. K. Ho and N. Armanfard, "Self-supervised learning for anomalous channel detection in EEG graphs: Application to seizure analysis," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 7, pp. 7866–7874.
- [54] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
- [55] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.
- [56] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
- [57] C.-C. Chiu, J. Qin, Y. Zhang, J. Yu, and Y. Wu, "Self-supervised learning with random-projection quantizer for speech recognition," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 3915–3924.
- [58] E. Eldele et al., "Time-series representation learning via temporal and contextual contrasting," 2021, *arXiv:2106.14112*.
- [59] Z. Yue et al., "Ts2vec: Towards universal representation of time series," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 8, pp. 8980–8987.
- [60] H. Xu, Y. Wang, S. Jian, Q. Liao, Y. Wang, and G. Pang, "Calibrated one-class classification for unsupervised time series anomaly detection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 5723–5736, Nov. 2024.
- [61] M. Tschannen, O. Bachem, and M. Lucic, "Recent advances in autoencoder-based representation learning," 2018, *arXiv:1812.05069*.
- [62] Y. Miao, L. Yu, and P. Blunsom, "Neural variational inference for text processing," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, vol. 48, Jun. 2016, pp. 1727–1736.
- [63] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Jozefowicz, and S. Bengio, "Generating sentences from a continuous space," 2015, *arXiv:1511.06349*.
- [64] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6341–6350.
- [65] C. Liu and J. Zhu, "Riemannian stein variational gradient descent for Bayesian inference," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.
- [66] A. Banerjee, I. S. Dhillon, J. Ghosh, S. Sra, and G. Ridgeway, "Clustering on the unit hypersphere using von Mises-Fisher distributions," *J. Mach. Learn. Res.*, vol. 6, no. 9, 2005.
- [67] R. A. Fisher, "Dispersion on a sphere," *Proc. Roy. Soc. London A, Math. Phys. Sci.*, vol. 217, no. 1130, pp. 295–305, 1953.
- [68] K. Guu, T. B. Hashimoto, Y. Oren, and P. Liang, "Generating sentences by editing prototypes," *Trans. Assoc. Comput. Linguistics*, vol. 6, pp. 437–450, Jul. 2018.
- [69] T. R. Davidson, L. Falorsi, N. De Cao, T. Kipf, and J. M. Tomczak, "Hyperspherical variational auto-encoders," 2018, *arXiv:1804.00891*.
- [70] H. Guan and W. A. P. Smith, "Structure-from-motion in spherical video using the von Mises-Fisher distribution," *IEEE Trans. Image Process.*, vol. 26, no. 2, pp. 711–723, Feb. 2017.
- [71] J. Xu and G. Durrett, "Spherical latent spaces for stable variational autoencoders," 2018, *arXiv:1808.10805*.
- [72] I. V. Serban, A. G. Ororbia II, J. Pineau, and A. Courville, "Piecewise latent variables for neural variational text processing," 2016, *arXiv:1612.00377*.
- [73] X. Chen et al., "Variational lossy autoencoder," 2016, *arXiv:1611.02731*.
- [74] S. Zhao, J. Song, and S. Ermon, "InfoVAE: Information maximizing variational autoencoders," 2017, *arXiv:1706.02262*.
- [75] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.
- [76] M. Abul Hasanat, J. Bohné, J. Milgram, S. Gentric, and L. Chen, "Von Mises-Fisher mixture model-based deep learning: Application to face verification," 2017, *arXiv:1706.04264*.
- [77] A. T. A. Wood, "Simulation of the von Mises Fisher distribution," *Commun. Statist.-Simul. Comput.*, vol. 23, no. 1, pp. 157–164, Jan. 1994.
- [78] G. Ulrich, "Computer generation of distributions on the M-sphere," *J. Roy. Stat. Soc., Ser. C*, vol. 33, no. 2, pp. 158–163, 1984.
- [79] C. M. Ahmed, V. R. Palleti, and A. P. Mathur, "WADI: A water distribution testbed for research in the design of secure cyber physical systems," in *Proc. 3rd Int. Workshop Cyber-Physical Syst. Smart Water Netw.*, 2017, pp. 25–28.
- [80] M.-L. Shyu, S.-C. Chen, K. Sarinnapakorn, and L. W. Chang, "A novel anomaly detection scheme based on principal component classifier," in *Proc. ICDM Found. New Direction Data Mining Workshop*, Piscataway, NJ, USA, 2003, pp. 172–179.
- [81] J. Audibert, P. Michiardi, F. Guyard, S. Marti, and M. A. Zuluaga, "USAD: Unsupervised anomaly detection on multivariate time series," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2020, pp. 3395–3404.
- [82] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.